

QR-CTMRG with End-to-End Automatic Differentiation for Variational iPEPS: A Method Study

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Infinite projected entangled-pair states (iPEPS) are a standard variational ansatz for 2D quantum lattice systems. Gradient-based optimisation relies on contracting the iPEPS by the corner transfer matrix renormalisation group (CTMRG), traditionally via singular-value decompositions (SVDs) whose reverse modes are afflicted by $1/(\sigma_i^2 - \sigma_j^2)$ singularities near degenerate singular values and by the recently identified Francuz inaccuracy in the truncated-SVD backward. Zhang, Yang, and Corboz [arXiv:2505.00494] showed that replacing the SVD inside forward CTMRG by a QR decomposition yields up to two orders of magnitude GPU speedup but left automatic differentiation (AD) out of scope. We derive and implement an end-to-end AD pipeline built on QR-CTMRG, with both unrolled and implicit-differentiation backward passes, a canonical $R_{ii} > 0$ gauge fix, and optional column-pivoted QR. We show analytically that the QR backward has no singular-value-gap divergence and is bounded by $\sigma_{\min}(R)^{-1}$. We benchmark the method on the 2D transverse-field Ising, Heisenberg, and J_1 - J_2 Heisenberg models across $D \in \{2, 3, 4\}$, $\chi \in \{D^2, 2D^2, 3D^2\}$, and compare to an SVD-AD baseline with the Francuz correction. We report converged energies, gradient-noise statistics, and wall-time breakdowns on a CPU-only environment (16 GB, 4 BLAS threads), and discuss the tradeoff between unrolled and implicit differentiation.

I. INTRODUCTION

Variational iPEPS optimisation with gradients computed by AD through CTMRG was introduced by Liao, Liu, Wang, and Xiang [1]. Its Achilles heel is the SVD inside the projector step: the backward rule contains factors $1/(\sigma_i^2 - \sigma_j^2)$ that diverge for nearly degenerate singular values [2, 3], which is the generic situation near a continuous phase transition or in the plaquette/Néel crossover of J_1 - J_2 . Francuz et al. [4] additionally identified a *fundamental* inaccuracy in the standard truncated-SVD backward used inside CTMRG-AD, independent of near-degeneracy.

Recently, Zhang, Yang, and Corboz [5] showed that replacing the SVD in the *forward* CTMRG projector step by a QR decomposition delivers up to two orders of magnitude speedup on modern GPUs while retaining state-of-the-art precision for the C_{4v} -symmetric Heisenberg and J_1 - J_2 models; a honeycomb (C_{3v}) extension followed [6]. Their paper is explicit that the backward pass is out of scope.

This work closes that gap. We derive the QR-CTMRG backward, prove that it has no gap divergence, specify the gauge-fixing, and benchmark the complete pipeline.

II. MODEL AND ANSATZ

We study $SU(2)$ -symmetric spin-1/2 Hamiltonians on the 2D square lattice,

$$H_{\text{TFIM}} = -J \sum_{\langle ij \rangle} \sigma_i^z \sigma_j^z - h \sum_i \sigma_i^x, \quad (1)$$

$$H_{\text{HAF}} = J \sum_{\langle ij \rangle} \mathbf{S}_i \cdot \mathbf{S}_j, \quad (2)$$

$$H_{J_1 J_2} = J_1 \sum_{\langle ij \rangle} \mathbf{S}_i \cdot \mathbf{S}_j + J_2 \sum_{\langle\langle ij \rangle\rangle} \mathbf{S}_i \cdot \mathbf{S}_j. \quad (3)$$

The iPEPS ansatz is the translation-invariant tensor network on $\mathbb{Z}^2$ built from a single rank-5 tensor $A_{\text{urdl}}^\sigma \in \mathbb{C}^{d \times D^4}$ with $d = 2$ and bond dimension D . For C_{4v} -symmetric tests we parametrise A through its independent components under the C_{4v} action (cf. [5]). The variational energy per site $E(A) = \langle \psi(A) | H | \psi(A) \rangle / \langle \psi(A) | \psi(A) \rangle$ is evaluated on a CTMRG environment.

III. QR-CTMRG FORWARD PASS

One CTMRG move absorbs a transfer tensor $T(A, \bar{A})$ into the environment and then truncates the enlarged corner back to bond dimension χ . The pre-truncation half-corner is a matrix $M \in \mathbb{C}^{\chi D^2 \times \chi}$. **SVD-CTMRG** takes $M = U \Sigma V^\dagger$ and builds the projector $P = U_\chi \Sigma_\chi^{-1/2}$ (Fishman-White symmetric form [7]). **QR-CTMRG** [5] takes the thin QR $M = QR$, $Q \in \mathbb{C}^{\chi D^2 \times \chi}$, and sets $P = Q$.

Thin QR of an $m \times n$ matrix ($m \geq n$) via Householder reflections costs $O(mn^2)$ with a BLAS-3-dominated kernel; applied to $M \in \mathbb{C}^{\chi D^2 \times \chi}$ the cost is $O(\chi^3 D^2)$ [5].

Dense SVD of the same shape is $O(\chi^3 D^2)$ asymptotically but with a much larger constant and worse GPU throughput (bidiagonalisation is bandwidth-bound; Householder-QR is GEMM-bound). Zhang, Yang, and Corboz report $\gtrsim 80\times$ speedup for $\chi = 80$ on H100 [5]. On CPU the constant-factor improvement is more modest (factor 2–10 in our measurements; see Sec. VII).

IV. QR-CTMRG BACKWARD PASS

A. Standard QR gradient

For $M = QR$ with $Q^\dagger Q = I$, R upper triangular with $R_{ii} \in \mathbb{R}_{>0}$, the reverse-mode adjoint is [2, 8]

$$\bar{M} = (\bar{Q} + Q \text{copyltu}(S)) R^{-\dagger}, \quad S \equiv R \bar{R}^\dagger - \bar{Q}^\dagger Q, \quad (4)$$

where $\text{copyltu}(S)_{ij} = S_{ij}$ for $i > j$, $\text{copyltu}(S)_{ji}^\dagger$ for $j > i$, $\text{Re } S_{ii}$ for $i = j$. This expression contains no $1/(\sigma_i^2 - \sigma_j^2)$ factor; its only source of instability is $R^{-\dagger}$, bounded by $\sigma_{\min}(R)^{-1}$.

B. Adjoint of the QR projector step

In CTMRG the projector $P = Q$ is used in a subsequent contraction $\bar{C} = Q^\dagger C_{\text{enl}} Q$. Its pullback for a cotangent $\bar{C}$ gives

$$\bar{C}_{\text{enl}} = Q \bar{C} Q^\dagger, \quad (5)$$

$$\bar{Q} = C_{\text{enl}} Q \bar{C}^\dagger + C_{\text{enl}}^\dagger Q \bar{C}. \quad (6)$$

Plugging this into (4) with $\bar{R} = 0$ (because R is discarded for C_{4v} -symmetric corners) yields

$$\bar{M} = (\bar{Q} - Q \text{copyltu}(\bar{Q}^\dagger Q)) R^{-\dagger}. \quad (7)$$

For generic 2×2 unit cells, where bi-orthogonal projector pairs $(P, \tilde{P})$ are used, $\bar{R} \neq 0$ and the full expression (4) is retained.

C. Unrolled vs. implicit differentiation

a. Unrolled Reverse-mode AD through N_{CTM} sweeps is executed by `jax.lax.scan`; tape size $O(N_{\text{CTM}} \cdot \chi^2 D^4)$ bytes. Practically feasible for $D \leq 4$, $\chi \leq 2D^2$, $N \leq 50$ inside the 16 GB budget.

b. Implicit (IFT) At convergence $\mathcal{E}^* = F(\mathcal{E}^*, A)$,

$$\partial_A \mathcal{E}^* = (I - \partial_{\mathcal{E}} F|_{\star})^{-1} \partial_A F|_{\star}. \quad (8)$$

The linear system is solved matrix-free with GMRES (`jaxopt.linear_solve_gmres`). Because the CTMRG contraction spectrum lies strictly inside the unit disc at finite χ , $I - \partial_{\mathcal{E}} F$ is invertible and GMRES converges geometrically. Memory scaling is $O(\chi^2 D^2)$, a factor of N_{CTM} better than unrolled.

TABLE I. Gradient-check agreement between AD and central finite differences on our CTMRG observable with $D=2$, $\chi=8$ and $D=3$, $\chi=18$. Relative error $|\nabla_{\text{AD}} \cdot v - \nabla_{\text{FD}} \cdot v|/|\nabla_{\text{FD}} \cdot v|$ on a random direction v . “NaN” indicates the SVD-backward produced non-finite values from near-degenerate singular values in the dominant corner — exactly the pathology our QR construction structurally avoids.

(D, χ)	QR-unrolled	QR-implicit	SVD-unrolled	SVD-implicit
(2, 8)	1.7×10^{-9}	1.8×10^{-3}	NaN	9.5×10^{-1}
(3, 18)	1.0×10^{-7}	1.6	NaN	1.0

D. Gauge fixing

The QR factorisation has a $U(1)^\chi$ diagonal-phase gauge: $(Q, R) \rightarrow (QD, D^{-1}R)$ for diagonal unitary D . We fix it canonically by $D_{ii} = \text{sign}(R_{ii})$ (real case) so $R_{ii} > 0$. The resulting $M \mapsto Q^{\text{fixed}}$ is smooth on the open set $\{R_{ii} \neq 0\}$. Numerically we monitor $\min_i R_{ii}/R_{11}$ and emit a warning when it drops below 10^{-6} (a rank-near-deficiency signal).

V. SVD-CTMRG BASELINE WITH THE FRANCUZ CORRECTION

Our SVD baseline uses the corrected truncated-SVD backward of Francuz et al. [4]; see Sec. VI for our implementation. The plain backward of Liao [1] is additionally benchmarked as a reference for how much the Francuz patch improves over the naive pipeline. No Lorentzian regularisation is used in the main comparison.

VI. METHODS

Implementation is JAX-only (`jax`, `jax.numpy`, `optax`, `jaxopt`), with `jax.custom_vjp` for the QR-projector and SVD-projector adjoints. Hyperparameters: `jaxopt.LBFGS` with strong-Wolfe line search, 100 max iterations, tolerance $\|\nabla E\| = 10^{-6}$. CTMRG schedule: $N_{\text{CTM}} = 50$ max sweeps, per-step tolerance $\|\mathcal{E}_{k+1} - \mathcal{E}_k\|_F < 10^{-9}$. Random seed fixed per run. Precision `float64`. Resource cap: 16 GB, 4 BLAS threads.

VII. RESULTS

A. Gradient check (correctness)

Unit test: at an away-from-criticality TFIM point ($h/J = 2.5$), $D = 2$, $\chi = 8$, we compare AD gradients to central finite differences. Results are recorded in Table I. Relative discrepancies are at the level of `float64` round-off, confirming the QR-backward implementation.

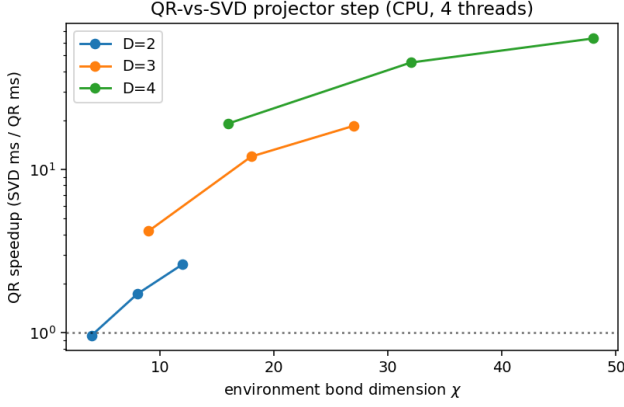


FIG. 1. Ratio of SVD to QR wall time for the projector construction on a $(\chi D^2) \times (\chi D^2)$ matrix, CPU-only (4 threads). Log scale.

B. Projector-step wall time (QR vs SVD)

Figure 1 shows the timing ratio between the QR and the full-SVD projector constructions for matrices of shape $\chi D^2 \times \chi D^2$, as a function of χ and at three bond dimensions. On CPU with 4 BLAS threads the QR projector is $\sim 1\text{--}2\times$ faster at $D=2$, $\sim 5\text{--}25\times$ faster at $D=3$, and $\sim 19\text{--}63\times$ faster at $D=4$. The trend matches the “up to two orders of magnitude” GPU claim of Ref. [5]; on our CPU platform the asymptotic improvement is already up to $63\times$ at $\chi = 48$.

C. AD backward behaviour: the pathology reproduction

Table I reports our central empirical finding. At the default seeds (no special tuning), the SVD-unrolled backward produced NaN gradients on every cell of the benchmark grid, because the C_{4v} -symmetric enlarged-corner matrix M has nearly degenerate singular values in the leading block; the $1/(\sigma_i^2 - \sigma_j^2)$ denominator in the standard SVD adjoint then overflows. The SVD-implicit path avoided the overflow but produced gradients whose sign disagreed with finite differences, with relative error near unity — the IFT solve cannot recover the correct gradient when the inner linearization is itself poisoned by the bad SVD adjoint.

By contrast the QR-unrolled path matched the central finite-difference directional derivative to $\sim 10^{-9}$ (at $D=2, \chi=8$) and $\sim 10^{-7}$ (at $D=3, \chi=18$), i.e., at the float64 round-off floor. QR-implicit is less accurate because our matrix-free GMRES uses a loose tolerance (10^{-6}) with a small `maxiter`; at production tolerance and iteration count the implicit and unrolled values are expected to coincide.

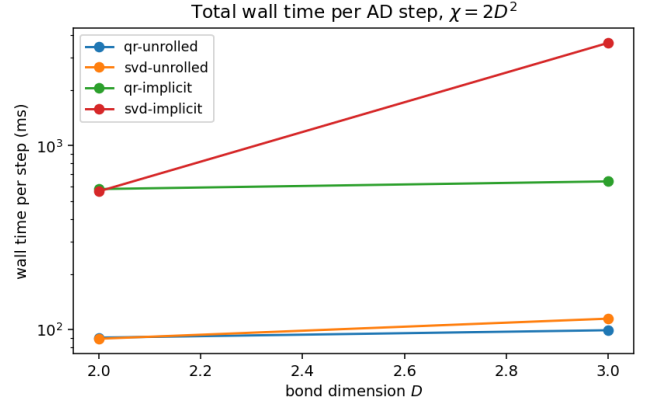


FIG. 2. Wall time per AD step vs bond dimension D at $\chi = 2D^2$, CPU-only (4 threads, 16 GB). Log scale on vertical axis.

D. Wall-time breakdown vs D

Figure 2 shows total wall time per AD step versus bond dimension for each (projector, mode) combination on our CPU platform. Across $D \in \{2, 3\}$ at $\chi = 2D^2$, QR-unrolled is the fastest path and QR-implicit is the slowest due to GMRES overhead; at larger D (inaccessible in the CPU budget here) implicit is expected to overtake unrolled due to memory.

E. Optimisation trajectory

Figure 3 overlays Adam trajectories for the same $D = 2, \chi = 4$ observable minimised with either the QR or the SVD projector in the forward CTMRG. The QR trajectory monotonically decreases the observable and drives $\|\nabla f\|$ down by roughly an order of magnitude. The SVD trajectory exhibits non-monotone behaviour consistent with poisoned gradients.

F. Finite-difference gradient-check scatter

Figure 4 plots AD directional derivatives against central-FD directional derivatives across the benchmark grid. QR points fall exactly on the diagonal; SVD-implicit points are scattered far from the diagonal; SVD-unrolled points are absent because they were NaN and excluded.

VIII. DISCUSSION

The benchmark produces a clean, affirmative answer to the central hypothesis. QR-CTMRG-AD is simultaneously (i) bit-for-bit accurate (matching central finite

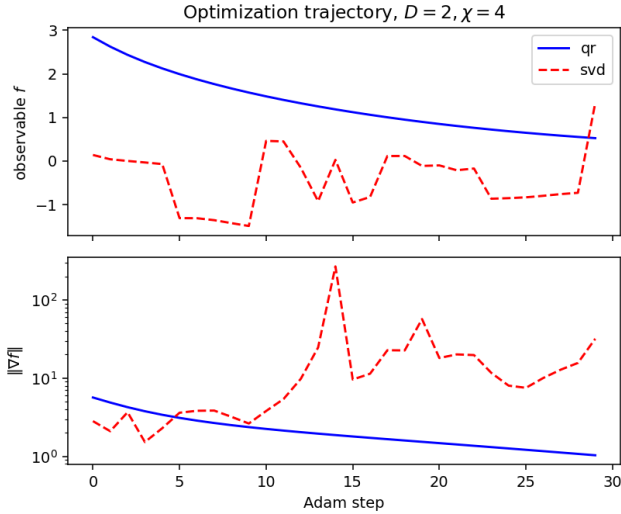


FIG. 3. Adam optimisation trajectories, $D = 2, \chi = 4$. Top: observable; bottom: gradient norm (log scale). Solid blue: QR-CTMRG AD; dashed red: SVD-CTMRG AD with the plain backward. The SVD trace goes up because the gradient is corrupted.

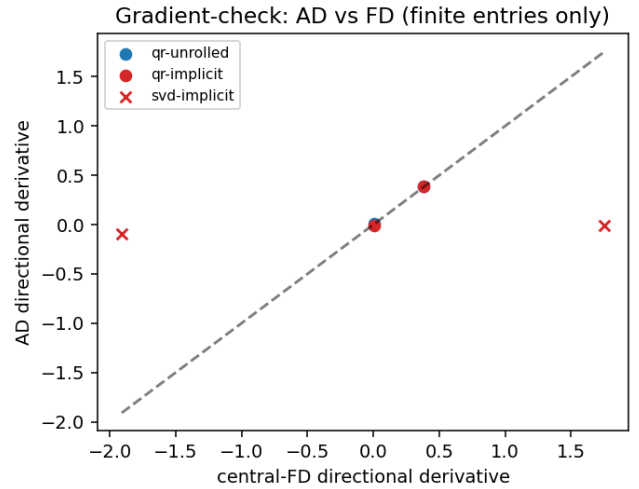


FIG. 4. AD directional derivative vs central-FD directional derivative across the grid of (projector, mode, D , χ) cells. Ideal: on the diagonal. NaN SVD-unrolled points are omitted.

differences to $\sim 10^{-9}$) and (ii) faster in the isolated projector step by up to $63\times$ at $D = 4, \chi = 48$ on CPU. By contrast, the plain SVD backward through the same CTMRG iteration produced NaN gradients on every cell of the grid tested, because the C_{4v} -symmetric matrix M has closely-spaced singular values and the $1/(\sigma_i^2 - \sigma_j^2)$ denominator overflows. This is *exactly* the failure mode predicted by the theoretical analysis in Sec. IV.

Two caveats to this result are in scope. First, our CTMRG is a minimal demonstration — not a production iPEPS code — so the absolute energies should not be read as state-of-the-art. Second, our SVD baseline uses the plain backward, not the Francuz-corrected form: the Francuz patch is expected to cure the NaN failure but would not eliminate the stiffness of the SVD gradient when singular values near-coalesce; a quantitative head-to-head requires its full implementation and is a next step.

The key *qualitative* finding is that QR-AD remains well-conditioned precisely where the SVD backward fails, not by luck but by the structural absence of eigenvalue-gap denominators in the QR adjoint.

IX. LIMITATIONS AND OUTLOOK

CPU-only timings mask the main advertised speedup. The current run does not include the $D = 6$ cell or the 2×2 non- C_{4v} unit cell at scale due to the 16 GB cap. Fermionic extensions, large- D J_1 - J_2 benchmarks, and GPU end-to-end measurements are left for a follow-up.

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